

Analysis Report

1. Introduction

The Anomaly-Based Accident Detection and Faulty Vehicle Identification System represents a transformative approach to traffic safety and incident management. This analysis report details the technical framework and operational logic of a hierarchical deep learning pipeline designed to automate the transition from real-time monitoring to forensic reporting. The system is structured to establish a behavioral baseline of normal traffic flow, utilizing spatio-temporal analysis to detect collisions and irregularities as anomalies. By integrating four specialized modules—object tracking, anomaly detection, video-based accident classification, and license plate recognition—the system provides a comprehensive diagnostic tool that minimizes human intervention. This report analyzes how these components interact to deliver objective evidence, fault attribution, and verified vehicle identities, serving as a roadmap for the development of a fully autonomous traffic investigation platform.

2. Current system

3. Proposed system

3.1 Overview

The proposed system is designed as a modular and hierarchical artificial intelligence pipeline that automates the entire process of accident detection and forensic analysis. The architecture is built on the principle of "continuous observation followed by triggered analysis." In the first layer, the system performs high-frequency object detection and multi-object tracking to digitize the trajectories of all vehicles in the frame. These trajectories are continuously evaluated by a spatio-temporal anomaly detection engine. When a collision or irregular movement is identified as an anomaly, the system transitions into an active diagnostic state. In this state, the system segments the relevant video sequence, classifies the specific type of accident, identifies the primary vehicle responsible for the incident through spatial correlation, and extracts the license plate of the liable actor. This modular approach ensures high computational efficiency by only activating complex diagnostic models when a potential incident is detected.

3.2 Functional Requirements

The proposed system's functionality is divided into four critical operational stages that work in a synchronized pipeline:

Continuous Object Tracking and Trajectory Mapping: The system must detect all vehicles within the camera's field of view and assign a unique identifier (ID) to each. It continuously records spatial coordinates to map the trajectory of every moving object.

Real-time Anomaly Triggering: Utilizing a spatio-temporal analysis, the system must distinguish between normal traffic flow and irregular incidents (collisions, sudden stops, or lane departures). It must trigger a system-wide alert when an anomaly score exceeds the confidence threshold.

Automated Event Extraction and Classification: Upon a trigger, the system must automatically isolate a 10-second window of the incident. It is required to classify the event into predefined categories such as "Rear-end Collision," "Side-swipe," or "Pedestrian Incident."

Causality-Based Fault Attribution and Identification: The system must analyze the interaction between vehicle trajectories at the moment of the anomaly to identify the "Primary Actor" (the vehicle that initiated the collision). Subsequently, it must perform high-precision License Plate Recognition (LPR) on the identified faulty vehicle to document legal identity.

3.3 Nonfunctional Requirements

Processing Latency and Real-Time Performance: The system must process video frames with minimal delay. The time elapsed from the detection of an anomaly to the generation of the final fault report should not exceed a few seconds to ensure near real-time operational efficiency.

Accuracy and Reliability (Pd vs. Fa): A critical requirement is maintaining a high Probability of Detection (Pd) for accidents while keeping the False Alarm rate (Fa) at an absolute minimum. The system must be robust enough to operate consistently under various lighting and weather conditions.

Scalability: The software architecture should be designed modularly to allow for the integration of additional camera streams or the upgrading of individual deep learning models without disrupting the entire pipeline.

Data Security and Integrity: Since the system processes sensitive information (license plates and accident clips), all stored data must be encrypted. Furthermore, incident logs must be tamper-proof to maintain their validity as forensic evidence.

Resource Efficiency: The pipeline must be optimized for high-performance computing, ensuring that the heavy workload of multiple concurrent models is managed effectively without system crashes or significant frame drops.

3.4 Pseudo requirements

Contextual Awareness: The system should be able to distinguish between a deliberate emergency stop (e.g., at a red light or stop sign) and an anomalous sudden halt caused by an accident, preventing false positives in structured traffic zones.

Temporal Coherence: When an anomaly is detected, the system must ensure the continuity of the vehicle ID; it should "remember" the pre-accident trajectory of a vehicle even if the collision causes a temporary occlusion or rapid change in shape.

Environmental Adaptability: The system should dynamically adjust its sensitivity thresholds for anomaly detection based on the time of day or known weather patterns to maintain consistent performance during night or rain.

Graceful Degradation: In the event of a partial system failure (e.g., if the LPR module cannot read a plate due to damage), the system should still finalize and save the incident report with the available trajectory and classification data rather than discarding the entire event.

Human-Centric Reporting: The final output should be structured in a way that a non-technical user (a police officer or insurance agent) can easily interpret the causality of the accident and the reason for the assigned fault.

3.5 System models

3.5.1 Scenarios

Scenario A: High-Speed Rear-End Collision

Context: Two vehicles are moving in the same lane; the leading vehicle slows down, and the following vehicle fails to brake.

System Response: The tracking module assigns IDs to both. Upon impact, the anomaly engine detects a sudden velocity change and spatial overlap. The system triggers the classification model, which labels it as a "Rear-End Collision." The fault logic identifies the following vehicle (ID: X) as the primary actor due to its pre-impact speed profile. The LPR module extracts the plate of vehicle X.

Scenario B: Intersection Violation (Red Light/Side Impact)

Context: A vehicle enters an intersection while the perpendicular traffic has the right of way, causing a T-bone collision.

System Response: The anomaly detector identifies an object entering a "prohibited zone" (based on learned traffic flow). Following the crash, the system analyzes the trajectory and determines that one vehicle violated the established flow. The incident is classified as a "Side Impact," and the plate of the violating vehicle is recorded.

Scenario C: Low-Visibility Night Incident

Context: An accident occurs during heavy rain or at night with limited street lighting.

System Response: The system utilizes infrared-optimized tracking (adjusted by its pseudo-requirements for environment adaptation). Although the image quality is lower, the spatio-temporal model focuses on the motion heatmaps to identify the anomaly. The system generates a report, noting the environmental constraints in the metadata.

3.5.2 Use case model

The system involves three primary actors: the Traffic Monitoring System (AI), the Administrator/Operator, and the Law Enforcement/Insurance Authorities.

UC1: Real-Time Traffic Surveillance:

Actor: AI System

Description: The system continuously tracks vehicle trajectories and maintains a normal behavioral baseline.

UC2: Incident Detection and Alerting:

Actor: AI System

Description: Upon detecting a spatio-temporal anomaly, the system triggers an alert and initiates the diagnostic pipeline.

UC3: Forensic Evidence Generation:

Actor: AI System

Description: The system classifies the accident, identifies the faulty actor, reads the license plate, and generates a structured report.

UC4: Data Review and Verification:

Actor: Administrator / Operator

Description: A human operator reviews the automated report and the 10-second clip to verify the AI's fault attribution.

UC5: Evidence Retrieval:

Actor: Law Enforcement / Insurance Authorities

Description: Authorized external entities access the encrypted database to retrieve verified accident reports and video evidence for legal proceedings.

3.5.3 Object and class model

The system is designed using an object-oriented approach where each physical entity and logical process is represented as a class. The core classes include:

Vehicle Class: Represents each vehicle detected in the frame. It stores attributes such as ID, CurrentPosition, VelocityVector, and TrajectoryHistory.

TrafficFrame Class: Acts as a container for all Vehicle objects in a single point in time, managing the metadata for the entire scene.

AnomalyDetector Class: A logical engine that analyzes TrajectoryHistory. It includes methods like calculateAnomalyScore() and triggerAlert().

Incident Class: Created only when an anomaly is triggered. It aggregates data from the Classifier, FaultAnalyzer, and LPRModule.

ReportGenerator Class: Handles the final output, formatting the data from an Incident object into a human-readable and secure forensic document.

3.5.4 Dynamic models

The dynamic behavior of the system is modeled through the interaction between the monitoring and diagnostic states. The process follows a strict event-driven sequence:

State Transitions: The system primarily resides in the "Observation State," where tracking and anomaly scoring are performed in a loop. Upon an anomaly trigger, it transitions to the "Diagnostic State," where parallel processes for classification and fault analysis are initialized.

Message Sequencing:

1. The AnomalyDetector sends a trigger signal to the BufferManager to lock the last 10 seconds of video.
2. The TrajectoryAnalyzer identifies the candidate Faulty ID and passes this metadata to the LPRModule.
3. The Classifier processes the video segment and returns the EventCategory.
4. All finalized data is pushed to the ReportGenerator to close the state and return to observation.

Concurrency: To maintain real-time performance, the LPR and Classification tasks are executed concurrently once the event clip is isolated.

3.5.5 User interface - navigational paths and screen mock-ups

The interface is designed as a Web-Based Dashboard focused on clarity and rapid response. The user flow is structured to guide the operator from a broad overview to a specific incident detail.

Navigational Paths:

Live View (Main): The entry point where real-time traffic streams are displayed with bounding boxes around tracked vehicles.

Incident Alert (Pop-up/Sidebar): A real-time notification that appears when an anomaly is triggered, leading directly to the classification results.

Archive/Report Library: A searchable repository where historical accident reports, clips, and plate data are stored.

Screen Mock-ups:

Dashboard Layout: A central video panel showing the "Anomaly Heatmap," a right-side panel listing "Current Active Vehicles," and a bottom notification bar for "Recent Alerts."

Incident Detail View: A split-screen layout showing the 10-second accident clip on one side and the automatically generated "Liability Table" (Faulty ID, Plate, Accident Type) on the other.

4. Glossary

Spatio-Temporal: Refers to data or analysis that relates to both space (positional coordinates) and time (sequential order), used to understand the progression of traffic movements.

Anomaly Detection: The technical process of identifying rare items, events, or observations which raise suspicions by differing significantly from the established "normal" traffic flow baseline.

LPR : License Plate Recognition; the technology used to detect the location of a vehicle's license plate within a video frame.

Trajectory: The continuous path followed by a moving vehicle through the camera's field of view, represented as a series of spatial coordinates over time.

Pd (Probability of Detection): A statistical performance metric representing the system's ability to correctly identify and trigger an alert for an actual accident.

Fa (False Alarm): An error metric where the system incorrectly identifies a normal traffic event or non-accident movement as an anomaly or collision.

Bounding Box: A rectangular coordinate frame used in computer vision to specify the exact location, height, and width of a detected vehicle in the video stream.

5. References

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